20. Project Execution, Local Setup, and Operational Runbook — Prescripto AI 2.0

Document Reference: `DOC-20-RUNBOOK`
Classification: Operational Runbook & Execution Manual
Repository Reference: `https://github.com/dipak0000812/Prescripto-2.0` (Branch `main`)
Deployment Backend: `https://prescripto-2-0.onrender.com`
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1. Quick Access and Service Topology

Subsystem	URL / Port	Technology	Purpose
Frontend Dev Server	`http://localhost:5173/`	React 18, Vite, Tailwind	Hot-reloading clinician and patient web portal with live API proxying.
Backend API Gateway	`http://localhost:8001/`	FastAPI, Uvicorn, Python 3.12	Core REST API gateway, authentication, and root SPA static file host.
Interactive API Documentation	`http://localhost:8001/docs`	Swagger UI / OpenAPI 3.0	Interactive testing harness for all 21 REST endpoints.
Alternative API Documentation	`http://localhost:8001/redoc`	ReDoc	Human-readable API specifications and schema models.
System Diagnostics	`http://localhost:8001/health`	FastAPI Health Router	Database pool, storage, and worker heartbeat checks.
Database Pool Diagnostics	`http://localhost:8001/health/db`	SQLAlchemy 2.0	PostgreSQL / SQLite active connection pool metrics.
Live Cloud Backend	`https://prescripto-2-0.onrender.com`	Render Docker PaaS	Publicly accessible production backend instance.

> ***Port Allocation Notice**: Port `8001` is utilized locally for the FastAPI backend because Windows Hyper-V / Docker dynamically reserves port `8000`. The Vite development configuration (`web/vite.config.ts`) automatically proxies `/api` requests to `http://127.0.0.1:8001`.*

2. Pre-Seeded Demonstration Accounts

The local database (`prescripto.db`) and cloud production databases are pre-populated with authenticated test roles:

2.1 Clinician / Reviewer Portal

- **Login URL**: `http://localhost:5173/login` (or `http://localhost:8001/login`)
- **Email Address**: `doctor.sharma@prescripto.local`
- **Username**: `doctor_sharma` *(or administrative user: `admin`)*
- **Password**: `DoctorPass123!` *(or `AdminPass123!`)*
- **Assigned Role**: `REVIEWER` (Grants access to dual-pane prescription review, bounding boxes, safety flags, and clinical sign-off).
- **One-Click Feature**: Use the **"Click to Autofill"** button directly on the login card.

2.2 Patient Mobile Voice Portal

- **Login URL**: `http://localhost:5173/patient-portal` (or toggle on login page)
- **Mobile Number**: Any valid 10-digit mobile number (e.g., `9876543210`)
- **Demo Verification OTP**: `123456`
- **Supported Languages**: English, Hindi (हिन्दी), Gujarati (ગુજરાતી), Marathi (मराठी) with client-side Web Speech API audio narration.

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3. Step-by-Step Local Startup Instructions

Prerequisites

- Python 3.11 or 3.12 installed on host machine.
- Node.js v20+ or v22+ and npm 10+.
- Microsoft Windows, macOS, or Linux.

Terminal 1: Launch Backend API Gateway

```
# Navigate to project root
cd c:\Users\pbhus\Desktop\Priscption-project\Prescripto-2.0

# Activate virtual environment and set Python module path
$env:PYTHONPATH="backend"

# Start Uvicorn ASGI server on port 8001
.\.venv\Scripts\python.exe -m uvicorn prescripto.api.main:app --host 127.0.0.1 --port 8001
```

Terminal 2: Launch Frontend Development Server

```
# Navigate to web application directory
cd c:\Users\pbhus\Desktop\Priscption-project\Prescripto-2.0\web

# Start Vite server with host exposure
npx vite --host 127.0.0.1 --port 5173
```

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4. Automated Verification and Test Execution

The system maintains 197 automated test cases verifying all modules. To re-run the verification suites locally:

```
$env:PYTHONPATH="backend"

# Execute Phase 1: Unit & Domain Logic Suite (68 tests)
.\.venv\Scripts\python.exe backend/scripts/run_phase1_api_suite.py

# Execute Phase 2: Database Schema & Migration Suite (24 tests)
.\.venv\Scripts\python.exe backend/scripts/run_phase2_database_suite.py

# Execute Phase 3: Integration & Pipeline Worker Suite (45 tests)
.\.venv\Scripts\python.exe backend/scripts/run_phase3_security_suite.py

# Execute Phase 4: Security, RBAC & DPDPA Deletion Suite (42 tests)
.\.venv\Scripts\python.exe backend/scripts/run_phase4_e2e_suite.py

# Execute Phase 5: Performance Benchmarks (18 tests)
.\.venv\Scripts\python.exe backend/scripts/run_phase5_performance_suite.py
```

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5. Live Cloud Deployment Synchronization

- **GitHub Repository**: `https://github.com/dipak0000812/Prescripto-2.0` (Branch `main`)
- **Render Auto-Deployment**: Render tracks `main` branch commits automatically.
- **Vercel Web App**: Connects to Render backend via `DEFAULT_API_BASE = 'https://prescripto-2-0.onrender.com'` configured in `web/src/api/client.ts`.